# Kittiphun Ruttanakorn



## Computer Technical Officer

Mahachulalongkornrajavidyalaya University, Phrae Campus


kittiphun.rut@mcu.ac.th | 089-955-4194 | Mueang Phrae, Phrae 54000, Thailand

kittiphun.rut.mcupr.ac.th | github.com/kittiphunr


## PROFESSIONAL SUMMARY


A computer technical officer with combined expertise in IT infrastructure and full-stack software development. I design, deploy, and maintain enterprise networks built on world-class equipment such as Cisco, administer Windows Server and Linux environments efficiently through the command-line interface, and build full-stack information systems that make the university's operations and services more reliable, secure, and modern. I also bring a background in industrial electronics and supervisory experience in manufacturing — leading electrical testing in R&D and machinery maintenance on production lines.


## CORE COMPETENCIES

**Enterprise Networking**

Cisco IOS / IOS-XE CLI · Routing & Switching · VLAN · Trunking · STP · OSPF · Static Routing · ACL & Network Security · Wireless LAN · Monitoring & Troubleshooting

**Windows Server**

Active Directory (AD DS) · DNS · DHCP · Group Policy · File & Print Services · Hyper-V · IIS · PowerShell

**Linux Server & CLI**

Ubuntu · Debian · RHEL-based · Bash Scripting · SSH & Hardening · Nginx · Apache · Docker · systemd · cron · Log Analysis

**Full-Stack Development**

HTML · CSS · JavaScript · Node.js · Python · REST API · SQL Databases · Git & GitHub · IoT (ESP32) · VBA Automation

## WORK EXPERIENCE

### Computer Technical Officer

**Oct 2024 – Present**

Mahachulalongkornrajavidyalaya University, Phrae Campus

- Design, configure, and maintain the campus network on Cisco equipment for high stability and security.
- Administer Windows Server and Linux servers, including user accounts, system services, security, and backups.
- Develop information systems and digital tools that support administration, procurement, and academic work.
- Manage the university's digital-media public relations.
- Provide technical consultation and support to staff and students.

### Freelance Full-Stack Developer

**2019 – 2023**

Self-employed

- Built full-stack web applications and information systems, covering the user interface, back end and database.
- Handled projects end to end — from gathering requirements and design through development to delivery.

**Head of Machinery Maintenance** 2017 – 2018

Thai An Pao Paper Products Co., Ltd. (Saha Group, Sriracha, Chonburi)

- Led maintenance of production machinery for paper cups and paper/plastic packaging.
- Production supplied packaging to Thai President Foods PCL.

**Head of Electrical Testing Laboratory, Research & Development** 2013 – 2017

Siam Compressor Industry Co., Ltd.

- Tested and developed projects on compressor packaging and installation under various operating conditions in refrigeration systems.
- Supervised the electrical testing laboratory within the R&D division.

## KEY PROJECTS

**Precision QC** JavaScript · Python · WebAssembly

Image-based quality control for fungal cultures

**8 analysis panels · 100% in-browser processing · ISO/IEC 17025 PDF report structure · 3 UI languages**

- Loads Pyodide (WebAssembly) to run the Python engine directly in the browser — no server, and images never leave the device
- Image processing with NumPy + Pillow: detects the dish with a Hough transform and converts pixels to centimetres

kittiphun.rut.mcupr.ac.th/projects/precision-qc-web

**Smart IV Alert** C++ · ESP32 · IoT · KiCad

IoT system for monitoring and alerting IV infusions (Smart IV Alert)

**28,000+ lines of C++ firmware · 8 beds per central unit · 1 s ESP-NOW reporting interval · 10/10 detector test scenarios passed**

- Schematics and PCBs in KiCad 7, generated from the pin numbers in the firmware so drawings and code never drift apart
- 28,000+ lines of C++ firmware across the bedside unit, three central-unit display variants and a flow simulator
- ESP-NOW, Wi-Fi + WebServer, mDNS, and Preferences (NVS) for persistent settings

github.com/kittiphunr/IV-Monitor-BCNPH

### Other Projects

- **TOR-MCU-Standard** — A toolkit for preparing Terms of Reference (TOR) documents according to university standards. (VBA, Automation)
- **BOQ-MCU** — A tool for preparing Bills of Quantities (BOQ) and construction cost estimates. (Python)
- **MCUPR-CANTEEN** — Canteen management system for MCU Phrae Campus. (C++, Embedded)
- **Music Player ESP32-S3** — A music player with a 2.8-inch display built on the ESP32-S3 microcontroller. (C, ESP32, Embedded)

## EDUCATION

**Bachelor of Technology (B.Tech.) in Electronics Technology**

Institute of Vocational Education: Northern Region 2

**Higher Vocational Certificate in Electronics (Industrial Electronics)**

Phrae Technical College

**Vocational Certificate in Electronics**

Phrae Technical College